

1/07: I carefully read through the article and decided to replicate Tables 1 and 2.

1/08: I identified the 16 variables required to replicate the paper and began planning how to divide the work across multiple R script files. I also revised my project timeline and reviewed the types of regression models used in the article to ensure I understood what I needed to do when I got to the analysis stage.

1/10: I loaded the data into R and filtered the dataset to include only observations from 2012.

When attempting to select the necessary variables, I discovered that two of the variables I needed were missing from the dataset, which prevented me from proceeding with the data cleaning/analysis as planned.

Files used:

mmarks9-PDIR-01-mgmt-setup.R

1/12: After discussing the missing variables with Dr. Bussberg, I learned that a different dataset was required. Once I received the correct dataset, I selected the relevant variables and filtered the data to match the sample size reported in the original study.

Files used:

mmarks9-PDIR-01-mgmt-setup.R

1/13: I created copies of variables that contained missing values and recoded “can’t choose” and “don’t know” responses as NA. I then performed multiple imputations using the mice package. Next, I reverse coded doearn and docskls. To prepare to take the descriptive statistics for Table 1, I also recoded sex into a binary variable and made edits to the income variable to make more values show up as missing to better match the original paper.

Files used:

mmarks9-PDIR-02-mgmt-missval.R (eventually had to change name/order)

mmarks9-PDIR-03-mgmt-impute.R (eventually had to change name/order)

mmarks9-PDIR-04-mgmt-revcode.R (eventually had to change name/order)

mmarks9-PDIR-05-mgmt-table1var.R (eventually had to change name/order)

1/14: I produced the descriptive statistics needed for Table 1 but realized that I shouldn't have performed the imputations until after generating these statistics because the "Can't choose"/ "No answer" responses needed to be included in the table. I adjusted the order of my workflow and got the descriptive statistics without imputed values. To take the statistics needed for table 1, I used the original versions of the variables before I coded the "can't choose" and "don't know" responses as NA. Additionally, the versions of docearn and docscls I used for the descriptive statistics were not reverse coded because taking the statistics of the normal versions of these variables better matched the paper's results. The missing values (with imputed new values) and the reverse coded doctor variables will be used in regression

Files used:

mmarks9-PDIR-02-mgmt-revcode.R

mmarks9-PDIR-03-mgmt-missval.R

mmarks9-PDIR-04-mgmt-table1var.R

mmarks9-PDIR-05-analysis-desc.R

mmarks9-PDIR-06-mgmt-impute.R

1/15: I constructed the multi-item physician distrust scale and attempted to standardize the scale for use in the regression models and a small part of Table 1. I then found the average, SD, and range of the scale- I knew these were off but wasn't sure how to fix them so I decided to come back to them later.

Files used:

mmarks9-PDIR-07-mgmt-scale.R

mmarks9-PDIR-08-analysis-table1scale.R

1/16: I created a template for Table 1 and filled it in with the descriptive statistics found in *mmarks9-PDIR-05-analysis-desc.R* and *mmarks9-PDIR-08-analysis-table1scale.R*

Files used:

mmarks9-PDIR-05-analysis-desc.R

mmarks9-PDIR-08-analysis-table1scale.R

1/17: I wrote and tested the code for the ordered logistic regression (OLR) and ordinary least squares (OLS) models. Because my scale was incorrect, my OLS models were pretty far from the original study. My OLR model was giving okay results, but it wasn't showing the p-values.

Files used:

mmarks9-PDIR-09-analysis-OLSreg.R

mmarks9-PDIR-10-analysis-OLRreg.R

1/19: After going to office hours, I learned that the physician distrust scale had been standardized incorrectly and rebuilt the scale accordingly. I corrected the descriptive statistics in Table 1, revised the OLS models, and completely changed my approach to estimating the OLR models by

switching from the *MASS* package to the *ordinal* package. I then created Table 2 and renamed all files to clearly distinguish between data management and analysis scripts.

Files used:

mmarks9-PDIR-07-mgmt-scale.R

mmarks9-PDIR-08-analysis-table1scale.R

mmarks9-PDIR-09-analysis-OLSreg.R

mmarks9-PDIR-10-analysis-OLRreg.R